Development of a Portable Fridge for Off Grid Vaccine Transport

Introduction

Malaria remains one of the world's most devastating diseases while still being preventable. There were an estimated **263 million cases** and **597,000 deaths** reported globally in 2023, with the vast majority of cases (**94%**) and deaths (**95%**) occurring in **Africa**. Malaria vaccines must be stored between **2–8℃** to maintain potency: Well below ambient temperature in most areas, especially Western Africa. Maintaining this cold chain requires **constant, reliable access** to electricity and specialized hardware. Many communities in Africa have **little or unreliable access** to electricity due to underdeveloped infrastructure.

One method to mitigate this issue is using a **Peltier module**. A Peltier module uses the thermoelectric effect to move heat from one face to the other when a voltage is applied. It is much **simpler** than usual pressurized refrigerant systems, but have been hindered by inefficiencies, though **recent improvements have been made.**

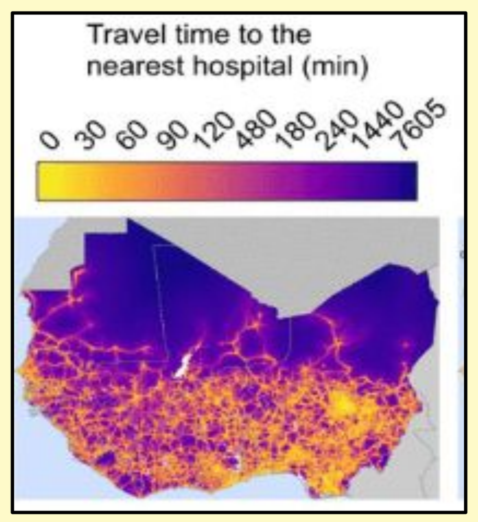

Fig. 1, Map of travel time to nearest hospital in West Africa. *Geldsetzer, P. et al., (2020). Mapping Physical Access to Healthcare for Older Adults in Sub-Saharan Africa: A Cross-Sectional Analysis with Implications for the COVID-19 Response.*

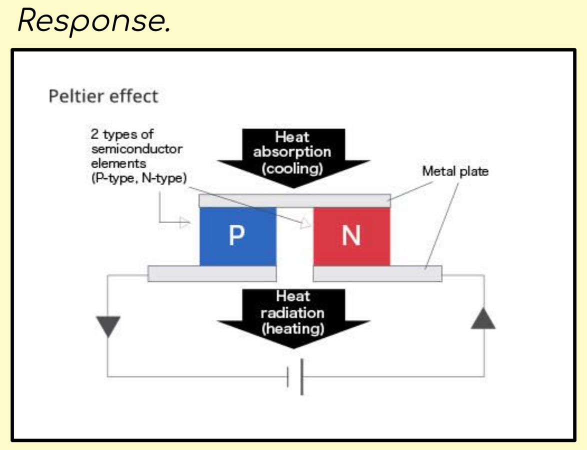

Fig 2, Peltier effect diagram. *Peltier Module (thermoelectric module) - energy conversion devices - Kyocera. KYOCERA GROUP GLOBAL SITE. (n.d).*

Purpose

There is a need for an off-grid, mobile refrigerator that can keep vaccines at the **2–8℃** temperature during transport **from distribution center to administration site.** An **increase in range** over previous solutions will allow the vaccines to travel much further from distribution centers, **expanding opportunities** to many rural communities who had not had access until now. Keeping **capital costs low** and **reusability** will increase effectiveness of funding.

Methods

1

- Physical
  - Cooler box (Igloo Ice Cube)
    - Simplify construction
  - Additional R-5 insulation boards
    - Added contact paper for protection and aesthetics
  - 3D printed trays
    - Hold vaccines still during transport

2

- Electronics
  - Raspberry Pi Zero 2W for control
    - Conserve energy by sensing temperature and adjusts peltiers accordingly
  - Dual channel relay, MPL3115A2 Thermometer
  - 2x7.4v 6200mAh LiPO batteries

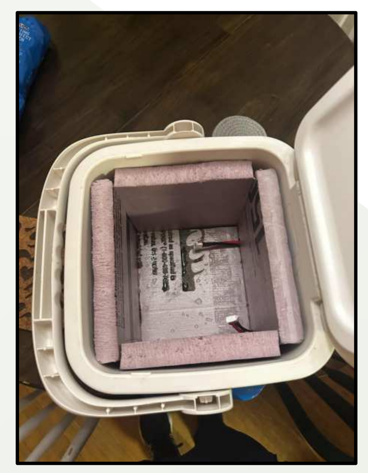
Fig. 3, Additional XPS insulation (Photo taken by finalist, 2026)

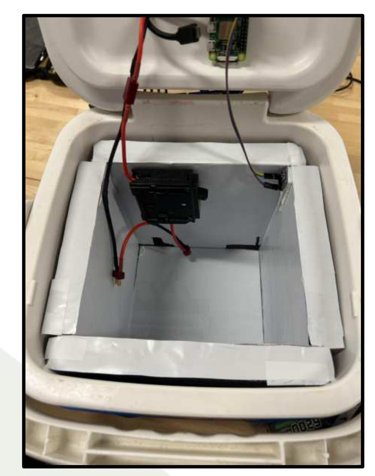
Fig. 4, Contact paper added (Photo taken by Finalist, 2026)

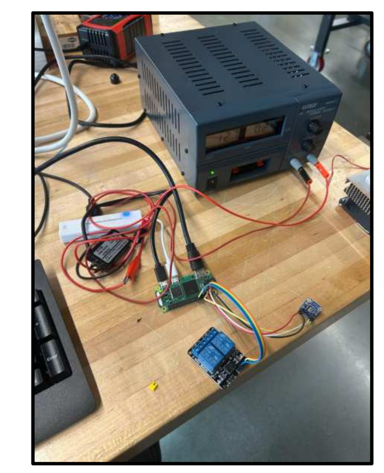
Fig 5, Electronics prototype (Photo taken by Finalist, 2026)


Fig 6, Mounted electronics (Photo taken by finalist, 2026)

3

- Software
  - Systemd Services on Pi
    - Periodically call python scripts that update current temperature, and change peltier state
    - ~1s intervals
    - Timestamped Temperature log kept.

4

- Testing
  - Tested overnight
  - Iced water used to simulate vaccines
  - Worst case scenario testing
    - Measured highest (warmest) air temperature
  - Initial testing without insulation
  - Final testing with added insulation and LCD Display

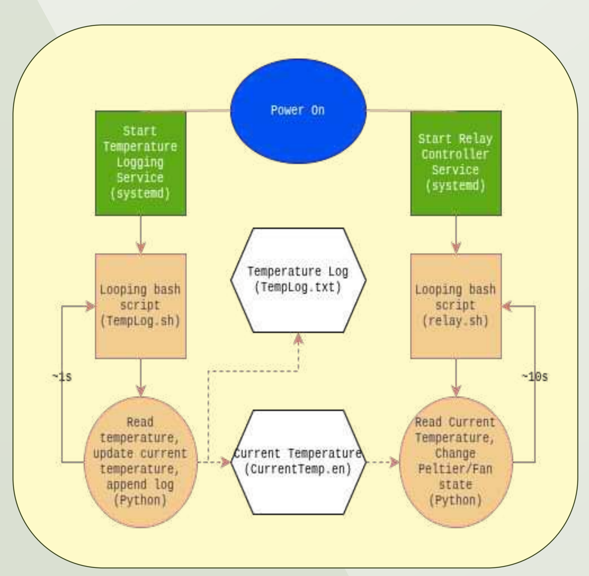

Fig 7, Software diagram (Diagram created by Finalist using Lucidchart, 2026)

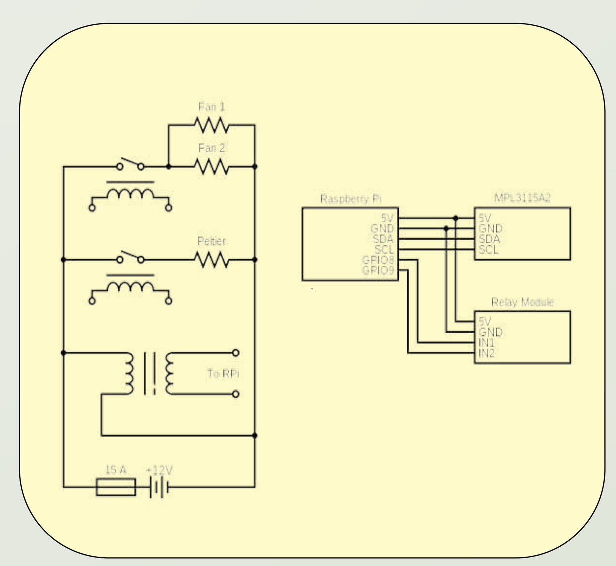
Fig 8, Wiring schematic (Schematic created by Finalist using Circuit-Diagram.org, 2026)

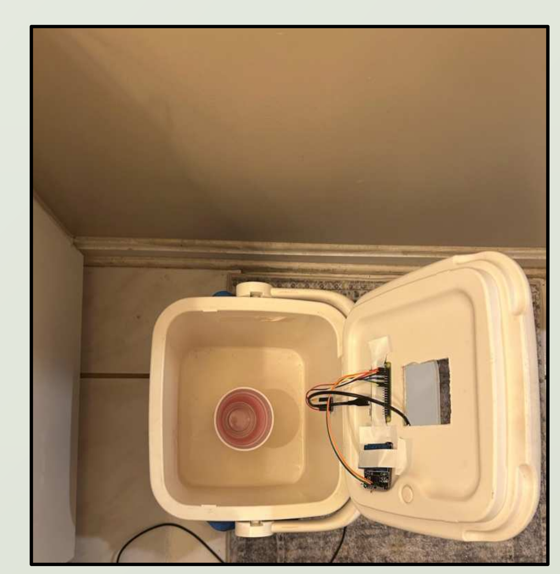
Fig. 9, initial testing setup before added insulation (Photo taken by Finalist, 2026)


Fig. 10, Final testing setup with added insulation (Photo taken by Finalist, 2026)

Data

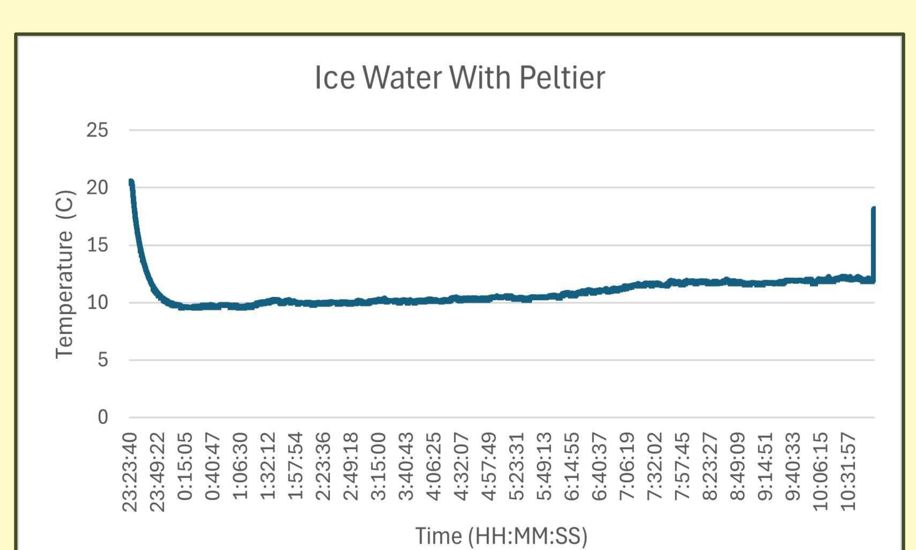

Fig 11, Temperature v. Time graph for Ice Water with Peltier (Graph created by Finalist, 2026)

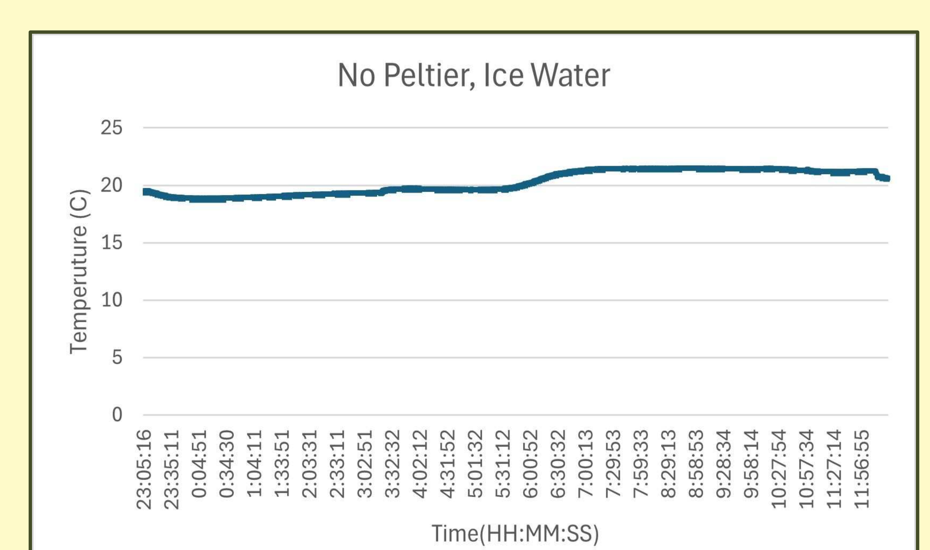

Fig 12, Temperature v. Time graph for Ice Water and No Peltier (Graph created by Finalist, 2026)

Conclusion

The thermoelectric refrigeration system developed successfully **maintained proper vaccine storage temperatures** over a continuous 24-hour period, demonstrating proper **thermal stability.** This indicates that Peltier-based cooling can meet **real-world medical storage requirements**, making it a viable solution for last-mile applications where reliable refrigeration and continuous access to management are limited.

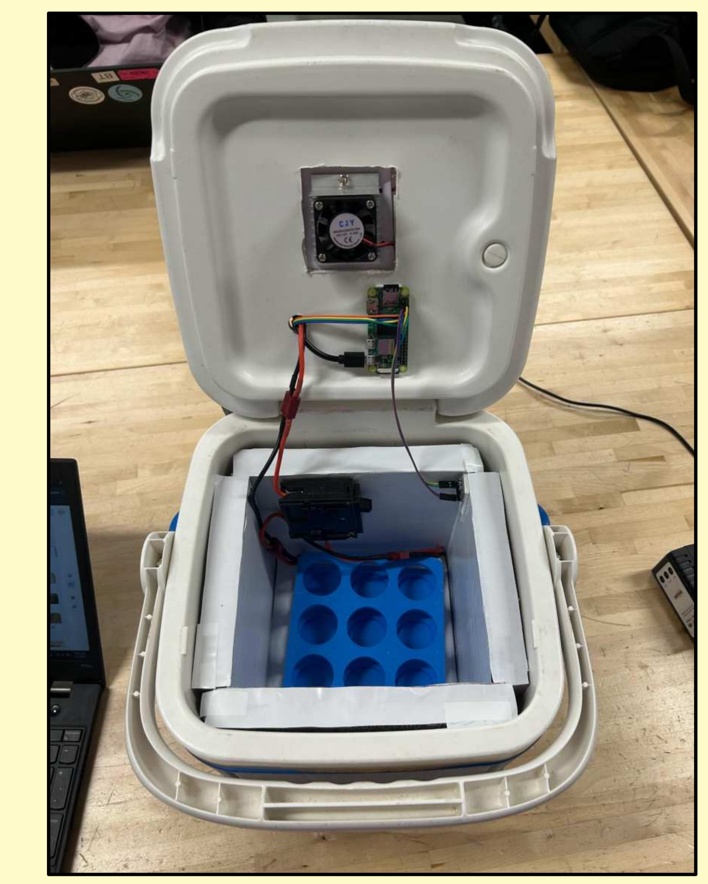
Fig 13, Completed system (Photo taken by Finalist, 2026)

Extensions

WHO's Performance and Quality standards for vaccine cool boxes are extremely close to being met with minor modifications. Vaccine storage capacity, hinge design, container shape, stackability and corrosion / chemical resistance requirements have already been met. **Full compliance can be achieved** through small design changes.

Future work should evaluate the system under conditions that can **mimic rural African environments,** including **higher ambient temperatures** and **minimal technical support**. Testing could be expanded to investigate **durability during shipping and handling**, as well as **longevity** of the electronics, particularly the batteries and Peltier.

Citations

Pambudi, N. A., Sarifudin, A., Gandidi, I. M., & Romadhon, R. (2022). Vaccine cold chain management and cold storage technology to address the challenges of vaccination programs. *Energy Reports*, *8*, 955–972. https://doi.org/10.1016/j.egyr.2021.12.039

Robertson, J., Franzel, L., & Maire, D. (2017). Innovations in cold chain equipment for immunization supply chains. *Vaccine*, *35*(17), 2252–2259. https://doi.org/10.1016/j.vaccine.2016.11.094

World Health Organization. (2023, October 2). *Malaria vaccines (RTS,S and R21)*. https://www.who.int/news-room/questions-and-answers/item/q-a-on-rts-s-malaria-vaccine

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